SIMATS ENGINEERING

CSA4712 – DEEP LEARNING
CO2 – ASSESSMENT 3

MATHEMATICAL ANALYSIS OF A DIABETIC RETINOPATHY SCREENING MODEL

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Assignment Overview

This assignment analyses the mathematical foundations of RetinaCare using the supplied data D1–D6. Each answer includes formulas, intermediate calculations, clear tables and interpretation. Information-theoretic quantities use logarithms to base 2. The design challenge considers the effect of 2% disease prevalence and a ten-to-one cost ratio for missed cases versus unnecessary referrals.

The question paper states that programming tools are not permitted; accordingly, the answers use mathematical derivations rather than code or simulated experiment screenshots.

TASK 1 — Break down the audit problem

The audit combines representation, optimization, uncertainty, clinical decision-making and evaluation. D1 is a feature matrix for singular value decomposition (SVD), rank and energy retention. D2 defines a quadratic loss surface; its Hessian determines curvature, convexity, conditioning and gradient-descent behaviour. D3 provides true and predicted class distributions for entropy, cross-entropy and KL divergence. D4 combines prevalence, sensitivity and specificity through Bayes' rule. D5 supports bias–variance analysis, while D6 provides the confusion matrix for classification metrics.

These items are connected: compression affects the representation, the loss and optimizer fit parameters, probability metrics describe output quality, and clinical metrics show the consequences of decisions. The calculations assume the supplied values are accurate and that sensitivity/specificity apply to the stated population. A sound optimizer does not guarantee a clinically safe classifier, so false negatives and prevalence must also be examined.

Data	Concept	Stage
D1	SVD, rank, matrix energy	Representation
D2	Hessian, eigenvalues, gradient descent	Loss / optimization
D3	Entropy, cross-entropy, KL	Probability quality
D4	Bayes' rule	Clinical decision
D5	Bias² + variance + noise	Prediction error
D6	Accuracy, precision, recall, F1	Evaluation

Analysis and Interpretation

Conclusion: no single result is sufficient. A compressed matrix may retain most energy but lose clinically useful information; a convex objective can converge while the model still misses referable disease. The audit must connect mathematical properties to the clinical cost of errors.

TASK 2 — Examine the compressibility of the feature matrix

Given A = [[3,0],[4,5]], compute A^TA = [[25,20],[20,25]]. Its characteristic equation is $\det(A^TA - \lambda I) = (25-\lambda)^2 - 400 = 0$, giving eigenvalues 45 and 5. The singular values are $\sigma_1 = \sqrt{45} = 6.7082$ and $\sigma_2 = \sqrt{5} = 2.2361$. The dominant singular value is three times the smaller one.

For $\lambda_1=45$, $v_1=(1/\sqrt{2})[1,1]^T$; for $\lambda_2=5$, $v_2=(1/\sqrt{2})[1,-1]^T$. The left singular vectors are $u_1=Av_1/\sigma_1=[3,9]^T/\sqrt{90}\approx[0.3162,0.9487]^T$ and $u_2=Av_2/\sigma_2=[3,-1]^T/\sqrt{10}\approx[0.9487,-0.3162]^T$. Thus $A=U\Sigma V^T$ with $\Sigma=\text{diag}(6.7082,2.2361)$.

Quantity	Calculation	Result
Eigenvalues of A^TA	45, 5	45 and 5
Singular values	$\sqrt{45}$, $\sqrt{5}$	6.7082, 2.2361
Energy	45+5	50
Rank-1 energy retained	45/50 × 100	90.0000%
Rank-1 residual norm	$\|\|A-A_1\|\|F=\sigma_2$	2.2361

Analysis and Interpretation

The rank-1 approximation is $A_1=\sigma_1u_1v_1^T$=[[1.5000,1.5000],[4.5000,4.5000]]. The residual is [[1.5000,−1.5000],[−0.5000,0.5000]], whose Frobenius norm is $\sqrt{5}$=2.2361. Rank-1 compression retains 90% of squared matrix energy. This is promising for compression, but the discarded 10% may still contain discriminative information; the compressed representation must be validated on downstream clinical classification.

TASK 3 — Analyze the curvature of the loss surface

$J(w)=\frac{1}{2}w^{T}Hw-b^{T}w$, where $H=[[4,1],[1,4]]$ and $b=[5,5]^{T}$. The characteristic equation is $\det(H-\lambda I)=(4-\lambda)^{2}-1=0$, so the eigenvalues are 5 and 3. Eigenvectors are $[1,1]^{T}$ and $[1,-1]^{T}$. Curvature is higher in the [1,1] direction (eigenvalue 5) and lower in the [1,−1] direction (eigenvalue 3).

Both eigenvalues are positive, so H is positive definite and J is strictly convex with a unique global minimizer. The condition number is $\kappa_{2}(H)=\lambda_{max}/\lambda_{min}=5/3=1.6667$, indicating modest curvature imbalance. Solving $Hw^{*}=b$ gives $w^{*}=[1,1]^{T}$.

Property	Derivation	Result
Eigenvalues	$(4-\lambda)^{2}-1=0$	5, 3
Positive definite	Both eigenvalues > 0	Strictly convex
Condition number	5/3	1.6667
Minimizer	$H^{-1}b$	$[1,1]^{T}$
Objective at w^{*}	$\frac{1}{2}(10)-10$	−5.0000

Analysis and Interpretation

Important consistency check: the brief asks to verify $J(w^{*})=0$, but the stated objective has no additive constant. Substitution gives $\frac{1}{2}w^{*T}Hw^{*}=5$ and $b^{T}w^{*}=10$, hence $J(w^{*})=-5$, not zero. The gradient is zero at the minimizer. A modified objective $J(w)=\frac{1}{2}w^{T}Hw-b^{T}w+5$ would have minimum value zero; the supplied formula should not be silently changed.

TASK 4 — Differentiate the models using information theory

The true distribution is $p=[0.5,0.25,0.125,0.125]$.

$H(p)=-\Sigma p_i \log_2 p_i = 0.5(1)+0.25(2)+0.125(3)+0.125(3)=1.7500$ bits. The uniform $q_1=[0.25,0.25,0.25,0.25]$ has entropy $H(q_1)=2.0000$ bits. Its cross-entropy with p is $H(p,q_1)=-\Sigma p_i \log_2(0.25)=2.0000$ bits.

Therefore $D_{KL}(p||q_1)=H(p,q_1)-H(p)=2-1.75=0.2500$ bits. For $q_2=[0.4,0.3,0.2,0.1]$, $D_{KL}(p||q_2)=\Sigma p_i \log_2(p_i/q_{2i})$. Class contributions are approximately 0.1610, −0.0652, −0.0847 and 0.0402 bits, summing to about 0.0513 bits.

Measure	Result (bits)	Interpretation
$H(p)$	1.7500	True-label uncertainty
$H(q_1)$	2.0000	Uniform prediction entropy
$H(p,q_1)$	2.0000	Cross-entropy
$D_{KL}(p\|\|q_1)$	0.2500	Untrained mismatch
$D_{KL}(p\|\|q_2)$	≈0.0513	Smaller mismatch

Analysis and Interpretation

q_2 is closer to p because its KL divergence is lower than q_1's. The largest positive class contribution is the first class, where $p=0.5$ but $q_2=0.4$. Some individual contributions can be negative, although total KL divergence cannot be negative. KL divergence zero means the distributions are identical over the classes. Better distribution alignment is encouraging, but calibration and per-class performance should also be examined.

TASK 5 — Analyze the screening decision using Bayes' rule

Prevalence $P(D)=0.10$, sensitivity $P(+|D)=0.90$ and specificity $P(-|\neg D)=0.95$. Therefore $P(+|\neg D)=0.05$ and $P(-|D)=0.10$. The total positive probability is $P(+)=0.90(0.10)+0.05(0.90)=0.135$.

Positive predictive value is $P(D|+)=0.09/0.135=0.6667$. Disease probability after a negative result is $P(D|-)=0.10(0.10)/[0.10(0.10)+0.95(0.90)]=0.01/0.865=0.0116$. Thus a positive test implies approximately 66.67% probability of disease under the stated assumptions, while a negative test leaves about 1.16% residual risk.

Expected outcome per 1,000	Calculation	Count
Patients with disease	0.10×1000	100
True positives	0.90×100	90
False negatives	0.10×100	10
Patients without disease	0.90×1000	900
True negatives	0.95×900	855
False positives	0.05×900	45

Analysis and Interpretation

Sensitivity is $P(+|D)$, not $P(D|+)$. Positive predictive value also depends on prevalence and false positives in the non-disease population. A positive screening result should lead to clinical assessment or confirmatory examination rather than automatic treatment, because screening identifies risk and does not independently establish a definitive diagnosis.

TASK 6 — Trace the convergence of gradient descent

The gradient is $\nabla J(w)=Hw-b$. With $\eta=0.1$ and $w_0=[0,0]^T$, the first gradient is $[-5,-5]^T$ and $w_1=[0.5,0.5]^T$. At w_1, the gradient is $[-2.5,-2.5]^T$, so $w_2=[0.75,0.75]^T$. At w_2, the gradient is $[-1.25,-1.25]^T$, so $w_3=[0.875,0.875]^T$.

Using the supplied objective, $J(w_1)=-3.7500$, $J(w_2)=-4.6875$ and $J(w_3)=-4.9219$. Distance to $w^*=[1,1]^T$ is 1.4142 at w_0, 0.7071 at w_1, 0.3536 at w_2 and 0.1768 at w_3. Distance halves each iteration because the update lies along the eigenvector with eigenvalue 5 and the error multiplier is $1-\eta\lambda=0.5$.

Iteration	w	J(w)	Distance to w*
0	[0.0000,0.0000]	0.0000	1.4142
1	[0.5000,0.5000]	−3.7500	0.7071
2	[0.7500,0.7500]	−4.6875	0.3536
3	[0.8750,0.8750]	−4.9219	0.1768

Analysis and Interpretation

The loss approaches −5, the minimum of the given formula; it does not approach zero because the source objective omits an additive constant. The steadily shrinking distance demonstrates convergence for $\eta=0.1$. Hessian eigenvalues determine how rapidly different directions converge, so the learning rate must remain within a stable range.

TASK 7 — Differentiate momentum from plain gradient descent

Use $v_t=\beta v_{t-1}+\nabla J(w_t)$ and $w_{t+1}=w_t-\eta v_t$, with $\beta=0.9$, $\eta=0.1$ and $v_0=w_0=[0,0]$. At the first step, $\nabla J(w_0)=[-5,-5]$, so $v_1=[-5,-5]$ and $w_1=[0.5,0.5]$. At w_1 the gradient is $[-2.5,-2.5]$; thus $v_2=0.9[-5,-5]+[-2.5,-2.5]=[-7,-7]$, giving $w_2=[1.2,1.2]$.

Plain gradient descent reaches [0.75,0.75] after two steps, whereas momentum reaches [1.2,1.2] and overshoots $w^*=[1,1]$. The accumulated velocity carries earlier movement into the current update, which can accelerate convergence but can also overshoot.

Method	After update 1	After update 2
Plain gradient descent	[0.5000,0.5000]	[0.7500,0.7500]
Momentum (β=0.9)	[0.5000,0.5000]	[1.2000,1.2000]

Analysis and Interpretation

For plain gradient descent on this positive-definite quadratic, the stability bound is $0<\eta<2/\lambda_{max}=2/5=0.4$. At $\eta=0.5$, the error multiplier along the $\lambda=5$ direction is $1-0.5(5)=-1.5$. Its magnitude exceeds one, so the error alternates sign and grows rather than converging. Momentum has a different stability analysis, so the standard bound is specifically the plain-gradient-descent bound.

TASK 8 — Decompose the prediction error

For $y=10$ and $\sigma^2=1$, Model P predictions [8,9,8,9] have mean 8.5, bias -1.5 and $\text{bias}^2=2.25$. Deviations from the mean are -0.5, 0.5, -0.5, 0.5; their squared average gives variance 0.25. Expected squared error is $\text{bias}^2+\text{variance}+\text{noise}=2.25+0.25+1=3.5000$.

Model Q predictions [12,7,11,8] have mean 9.5, bias -0.5 and $\text{bias}^2=0.25$. Deviations are 2.5, -2.5, 1.5, -1.5, giving variance $(6.25+6.25+2.25+2.25)/4=4.25$. Its expected squared error is $0.25+4.25+1=5.5000$.

Model	Mean	Bias²	Variance	Expected squared error
P	8.5000	2.2500	0.2500	3.5000
Q	9.5000	0.2500	4.2500	5.5000

Analysis and Interpretation

Model P is preferable because its expected squared error is lower. Its dominant issue is systematic bias, since predictions consistently fall below the target. Model Q has less bias but much larger variance, so predictions are less stable. The empirical mean squared residuals of the listed predictions are 2.5000 for P and 4.5000 for Q; these differ from expected error because the requested decomposition adds the assumed irreducible noise variance.

TASK 9 — Examine the error profile of the classifier

Given TP=40, FN=10, FP=5 and TN=45, there are 100 images. Accuracy=(TP+TN)/N=85/100=0.8500, while error rate=(FP+FN)/N=15/100=0.1500. Precision=TP/(TP+FP)=40/45=0.8889. Recall or sensitivity=TP/(TP+FN)=40/50=0.8000. Specificity=TN/(TN+FP)=45/50=0.9000.

F1=2TP/(2TP+FP+FN)=80/(80+5+10)=0.8421. There are ten false negatives and five false positives. In diabetic-retinopathy screening, false negatives are especially concerning because a patient needing referral may be missed; false positives can cause unnecessary referral, anxiety and workload.

Metric	Formula	Result
Accuracy	(TP+TN)/N	85.0000%
Error rate	(FP+FN)/N	15.0000%
Precision	TP/(TP+FP)	88.8889%
Recall / sensitivity	TP/(TP+FN)	80.0000%
Specificity	TN/(TN+FP)	90.0000%
F1-score	2TP/(2TP+FP+FN)	84.2105%

Analysis and Interpretation

Accuracy alone hides the different consequences of false negatives and false positives. Recall of 80% means one in five actual referable cases in this sample is missed. A clinical audit should report the confusion matrix, recall, specificity, precision and F1 together, and choose the operating threshold using prevalence and the relative cost of errors.

TASK 10 — Relate the findings to the audit question

Task 2 found singular values 6.7082 and 2.2361; rank-1 approximation retains 90% of matrix energy. This supports compression as a candidate, but does not prove that disease classification will be preserved. Task 3 found positive Hessian eigenvalues 5 and 3 and condition number 1.6667, indicating a strictly convex and reasonably conditioned objective. Task 6 shows convergence at $\eta=0.1$, with distance to the minimizer halving each iteration.

Task 4 shows q_2 is closer to the true distribution than q_1: KL divergence decreases from 0.2500 bits to about 0.0513 bits. However, Task 9 shows recall of 80% and ten false negatives out of 50 actual referable cases. These misses are a central concern because delayed referral can cause preventable visual harm.

Audit question	Evidence	Conclusion
Compression	90% matrix energy retained	Promising; validate task performance
Optimization	H positive definite; $\kappa=1.6667$	Stable with suitable learning rate
Probability quality	KL: 0.2500 to ≈0.0513 bits	q_2 aligns better with p
Clinical errors	Recall 80%; 10 false negatives	Missed cases need attention
Population change	PPV depends on prevalence	Recalculate locally

Analysis and Interpretation

The mathematical components are interpretable, but approval should not rely on compression or convergence alone. A screening policy needs explicit clinical cost analysis and threshold review. Task 11 shows that low prevalence greatly reduces positive predictive value even when sensitivity and specificity stay constant, so the policy should be modified for the new district and validated on representative local data.

TASK 11 — Design challenge: impact of low prevalence

At 2% prevalence, P(D)=0.02 and P(¬D)=0.98. True-positive probability is 0.02×0.90=0.018; false-positive probability is 0.98×0.05=0.049, so P(+)=0.067. Therefore P(D|+)=0.018/0.067=0.2687. For a negative result, P(D|−)=0.02×0.10/[0.02×0.10+0.98×0.95]=0.002/0.933=0.0021.

Sensitivity and specificity remain unchanged by assumption, but post-test probabilities change because prevalence is the prior probability. Positive predictive value falls from 66.67% at 10% prevalence to 26.87% at 2%. Per 1,000 patients at 2% prevalence, expected outcomes are TP=18, FN=2, TN=931 and FP=49.

Expected outcomes per 1,000	10% prevalence	2% prevalence
True positives	90	18
False negatives	10	2
False positives	45	49
True negatives	855	931
P(Disease | positive)	66.6667%	26.8657%
P(Disease | negative)	1.1561%	0.2144%

Analysis and Interpretation

For PPV ≥0.5 at 2% prevalence, 0.018/[0.018+0.98(1−specificity)]≥0.5. Rearranging gives specificity ≥1−0.018/0.98=0.9816, or 98.1633%. If sensitivity remains 90%, this corresponds to about 18 false referrals and 2 missed cases per 1,000. Using cost 10 per missed case and 1 per unnecessary referral, the current policy costs 10×10+45=145 units at 10% prevalence and 10×2+49=69 units at 2% prevalence. At the higher specificity, illustrative cost is about 10×2+18=38 units per 1,000.

Evidence-Based Recommendation

RECOMMENDATION: MODIFY the policy. The ten-to-one cost ratio means missed cases deserve strong priority, while improving specificity can reduce unnecessary referrals. In real threshold selection, sensitivity and specificity usually change together, so the exact cost of another threshold cannot be calculated without its ROC operating point. Validate the chosen threshold on representative local data, retain confirmatory clinical review, and consider rural referral capacity and follow-up access.

Overall Conclusion

The calculations support mathematical review, but clinical approval requires more than accuracy or optimizer convergence. Low prevalence lowers positive predictive value, while false negatives carry high cost. The screening policy should be modified and locally validated, with sensitivity and specificity monitored and confirmatory clinical referral pathways preserved.